AI-based Damage Detection for Power Grid Equipment with SaaS-oriented Design

Project Closing Report

Team Leader: Zhai Zhengpeng
Members: Feng Yujie, Lu Yuhao

SWJTU-Leeds Joint School
Southwest Jiaotong University

December 23, 2025

Executive Summary: From “Experimental” to “Industrial”

Original Scope

- Generative Augmentation (Stable Diffusion/LoRA) for Transformer faults.
- Experimental Architecture (WHRT-DETR).

Execution Strategy (The Pivot)

- **Focus:** Transmission Line Bird Nest Detection.
- **Goal:** A high-fidelity “Vertical Slice” SaaS product.
- **Rationale:**
  - **Data Availability:** Transformer internal data is proprietary; transmission lines are visually accessible.
  - **Reliability:** Generative methods proved unstable (“hallucinations”).
  - **Verification:** Visual faults allow for rigorous PoC validation.

Methodology (1): Robust Data Pipeline

The “Generative Trap” & The Deterministic Solution

Issue

Generative models introduced physically implausible artifacts (e.g., rust on non-metallic parts).

Solution: Deterministic Augmentation

Implemented using `torchvision.transforms.v2`:

- **Custom Random IoU Crop:** Forces model to learn from partial object views (Scales: 0.3–1.0).
- **Photometric Distortions:** Simulates field conditions (Brightness $\pm 12.5\%$, Contrast $\pm 50\%$).
- **Geometric Regularization:** Consistent aspect ratios via `ResizeMax` and `PadSquare`.

Methodology (2): Model Architecture

Configuration for Stability & Speed

Model: Mask R-CNN (ResNet50-FPN v2)

- *Why?* Proven baseline for instance segmentation.

Key Stats:

- Input: 512×512
- Epochs: 40
- Batch Size: 4

Training Hyperparameters:

Parameter	Value
Optimizer	AdamW (wd: 0.01)
Scheduler	OneCycleLR
Peak LR	5×10^{-4}
Precision	Mixed (AMP)

PoC Progress: Training Dynamics

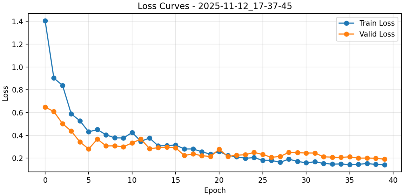

Figure: Train/Validation Loss

Key Observations

- Loss drops significantly and stabilizes around epoch 20.
- Train Loss: $1.41 \rightarrow 0.14$
- Val Loss: $0.65 \rightarrow 0.19$
- No significant overfitting observed with current augmentation.

Results: Quantitative Performance

Validation Metrics

Metric	Score
Box AP_{50}	1.000
Mask AP_{50}	1.000
Mask AP_{75}	0.919

Perfect scores at AP_{50} indicate reliable detection; AP_{75} confirms precise segmentation masks.

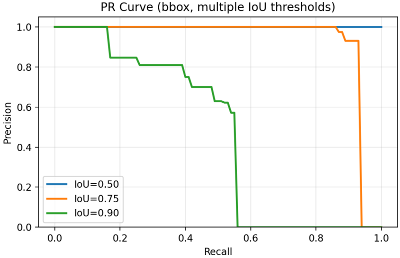

Figure: PR Curve (Bbox)

Results: Precision & Sensitivity Analysis

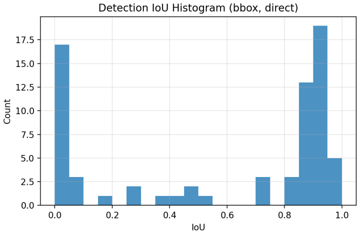

Figure: Bbox IoU Histogram

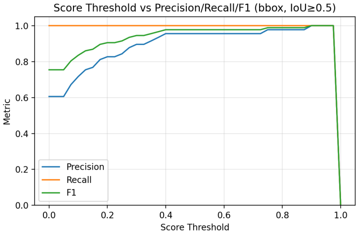

Figure: Threshold vs P/R/F1

Right-skewed IoU mass indicates strong localization.
Thresholds 0.4–0.5 optimize F1 score.

SaaS Deployment: Production-Ready Microservice

Architecture Highlights

- **Ingestion:** Batch upload support.
- **Inference:** Real-time GPU processing.
- **Output:** JSON metadata + Visual overlay.

Platform Significance

- Validates "Model-as-a-Service".
- Scalable multi-tenant design.
- User-friendly for field operators.

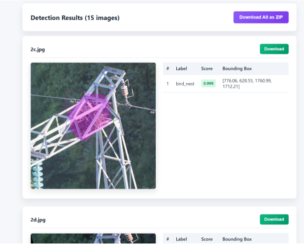

Figure: Batch Processing Interface

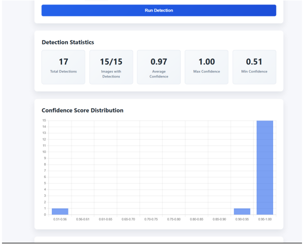

Figure: Inference Visualization

Discussion: Key Engineering Decisions

Mask R-CNN vs. YOLO

- **Why Mask R-CNN?** Instance Segmentation is critical for damage assessment (pixel-level).
- **Trade-off:** 100-200ms latency is acceptable for post-flight analysis.

Generative vs. Deterministic

- **Failure Mode:** Stable Diffusion generated "noise patterns" instead of semantic nest features (Feature Leakage).
- **Pivot:** Deterministic augmentation ensures physical fidelity and traceability.

Conclusion

Deliverables

- Vertical Slice: Data $\rightarrow$ Training $\rightarrow$ SaaS $\rightarrow$ UI.
- AP_{50} 1.0 on Bird Nest Module.
- Robust Flask microservice.

Impact

- Validated SaaS-oriented design.
- Blueprint for future Oil/Infrared modules.

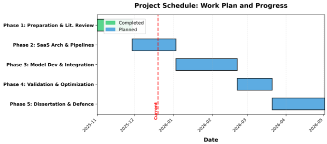

Figure: Project Timeline

Thank You

Questions?

Discussion: Multi-modal integration, Scalability, Edge deployment